

Part I: Measurement Cost of Generator Selection in Fixed-State ADAPT-VQE

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Abstract

This report defines and compares three measurement strategies for the generator-selection step in gradient-based ADAPT-VQE (adaptive derivative-assembled pseudo-Trotter variational quantum eigensolver): **M1: FC-UG**, fully commuting (FC) universal groups for static all-gradient estimation; **M2: BAI-FC-IG**, best-arm identification (BAI) using FC individual-gradient measurements; and **M3: BAI-FC-UG**, best-arm identification using fixed parent FC universal groups. A fourth construction, a *no-sharing baseline* that uses neither mechanism, completes the two-by-two the three methods span and makes each mechanism separately attributable.

Two classes of number are reported, and they are kept apart throughout. *Planning bounds* (Secs. 9–10) are oracle diagnostics: exact gradients and exact fragment variances are used to compute the shots each method would need, and the elimination rule reads exact values, so it cannot misfire. A *finite-shot surrogate* (Sec. 11) replaces the exact gradients with Gaussian estimates built from a finite number of shots and runs the same elimination rule on those. The surrogate samples estimators, not individual measurement outcomes, so it is not a shot-level simulation of a quantum device; the distinction and its consequences are set out in Sec. 7.

Results. On the planning bounds, M3 is cheaper than M1 in all nine cases, by factors of 1.1 to 192, and cheaper than M2 in eight of nine. The M2-versus-M3 comparison is *unresolved* on every H₂O row: the ratio moves over 0.74 to 1.30 across arbitrary choices of grouping routine, which spans unity. Against the no-sharing baseline, elimination alone is worth 1.3 to 253 and sharing alone only 1.1 to 3.4, and the two mechanisms are substitutes rather than complements: the interaction term is below unity in all nine cases. Under the finite-shot surrogate both best-arm methods cost more than their bounds, M2 by 2.2 to 3.2 and M3 by 1.4 to 3.0; an unseparated part of that is discretisation of the radius schedule rather than noise, and no mechanism for the difference between the two methods is established. Because M1 realises its bound exactly, the comparisons against M1 reorder under finite shots and M3 loses on two of seven rows. Every trial of the main finite-shot table returned the correct generator, which at 200 trials bounds the error rate only below 1.9%; the stated family-wise failure probability does not control that rate.

Scope. Every number is a single-ADAPT-step diagnostic for one fixed state, and counts shots only: circuit depth, the number of distinct measurement settings, and state-preparation cost are excluded (Sec. 7). Part III symmetry filtering and tapering are not included in the Part I tables; Part III supplies a smaller input problem to the same measurement methods. The revision history of this document, the naming convention for code and files, and the implementation status are in Apps. A–B.

1 Problem statement and relation to previous work

Given a molecular Hamiltonian H , a current ADAPT-VQE state $|\psi\rangle$, and a unitary coupled-cluster singles-and-doubles (UCCSD) generator pool $\{G_i\}$, estimate the measurement cost required to identify a generator with the largest absolute gradient

$$i^* = \arg \max_i |g_i|, \quad g_i = \langle \psi | [H, G_i] | \psi \rangle. \quad (1)$$

The scientific question is whether using winner-identification statistics and shared fully commuting measurement contexts reduces the gradient-selection measurement cost relative to static all-gradient estimation.

The answer, in one sentence. Mostly yes: on oracle planning bounds at family-wise $\delta = 0.05$, combining both mechanisms (M3) costs 1.1 to 192 times less than static all-gradient estimation (M1) across the nine cases studied — but the change of statistical target carries the saving in seven of the nine, shared contexts being worth only 1.1 to 3.4 alone; the advantage shrinks under finite shots and reverses on two rows; and the comparison between the two best-arm variants is unresolved on the largest system.

Here g_i is the derivative of the energy with respect to the variational parameter of generator G_i at $\theta_i = 0$, in Hartree per radian; G_i is anti-Hermitian, so g_i is real. The generator pool and its labelling convention are fixed in Sec. 6.

Relation to previous work. ADAPT-VQE and the spin-orbital UCCSD pool used here are those of Ref. [1]. Two lines of work attack the selection-step measurement cost separately. Ref. [3] attacks it by *sharing*: it partitions the gradient observables into commuting sets using Hamiltonian terms as pivots, reducing the pool-gradient cost to a small multiple of one energy measurement, and Ref. [2] pursues a related reduction alongside symmetry considerations. Ref. [4] attacks it by *elimination*: it recasts generator selection as best-arm identification and applies successive elimination, discarding unpromising candidates early. Ref. [5] changes the selection criterion itself.

The two mechanisms have not, to our knowledge, been measured against each other on one cost criterion, and it is not obvious that they compose. This report supplies the missing comparison rather than a new algorithm. Its contributions are: the 2×2 factorial of sharing and elimination, completed by a no-sharing baseline so that each mechanism and their *interaction* are separately attributable (Sec. 10); a single shot criterion with no free schedule or precision parameter, applied identically to all four constructions; and the finding that the two mechanisms are substitutes rather than complements, with elimination carrying the result. M1 stands in for the sharing line, M2 for the elimination line, and M3 for their combination.

2 Fixed method names

The phrase “naive/no-sharing” is retired as a *method* name. A construction that shares nothing does appear below, as the fourth corner of the two-by-two that the three methods span; it is called the *no-sharing baseline* throughout, it is a diagnostic rather than a project method, and it is defined in Sec. 10. App. B fixes the naming convention for code and files.

M1: FC-UG, Fully Commuting Universal Groups. One global parent FC grouping of the union of all Pauli products in all gradient commutators; estimate the full gradient vector.

M2: BAI-FC-IG, BAI with FC Individual Gradients. Each generator gradient is a BAI arm, measured in an FC grouping of its own commutator. No cross-gradient sharing.

M3: BAI-FC-UG, BAI with FC Universal Groups. The same fixed parent contexts as M1, with BAI elimination, sampling only the parent contexts needed to separate surviving generators.

No-sharing baseline. A diagnostic, not a project method: each generator grouped separately as in M2, every one resolved to the common radius used by M1 (Sec. 10).

3 Object lineage

The qubit Hamiltonian is

$$H = \sum_p h_p P_p, \quad (2)$$

where P_p are Pauli products after Jordan–Wigner mapping. For each generator G_i , define the gradient operator

$$C_i = [H, G_i]. \quad (3)$$

The ADAPT gradient is the expectation value of this operator,

$$g_i = \langle C_i \rangle_\psi. \quad (4)$$

Expanding all gradient operators in one shared Pauli-coordinate system gives

$$C_i = \sum_\ell A_{i\ell} R_\ell, \quad \mu_\ell = \langle R_\ell \rangle_\psi, \quad \mathbf{g} = A\boldsymbol{\mu}. \quad (5)$$

The parent universal Pauli support at this ADAPT step is

$$\mathcal{B}_0 = \bigcup_i \text{supp}(C_i). \quad (6)$$

M1 and M3 use $\mathcal{B}_0$ as a shared measurement object. M2 does not; it groups each C_i separately.

4 The three algorithms

4.1 M1: FC-UG

M1 estimates the full gradient vector. It forms one global FC grouping of the parent support,

$$\mathcal{B}_0 \longrightarrow \mathcal{M}_1, \dots, \mathcal{M}_K, \quad (7)$$

where each $\mathcal{M}_\alpha$ is a fully commuting measurement context. Measurements of these contexts estimate the shared Pauli expectations μ_ℓ and reconstruct every gradient using Eq. (5).

The implementation uses deterministic greedy FC grouping of $\mathcal{B}_0$ (Sec. 6). It is a generic stand-in for the pool-wide grouping strategies of Refs. [3, 2] and is not claimed to reproduce either construction; Sec. 13 shows how much the choice of routine matters.

4.2 M2: BAI-FC-IG

M2 changes the statistical target from full-vector estimation to winner identification. Each candidate gradient is an arm in a best-arm-identification problem. Its own commutator support is grouped separately,

$$\text{supp}(C_i) \longrightarrow \mathcal{M}_{i,1}, \dots, \mathcal{M}_{i,K_i}. \quad (8)$$

Throughout, t indexes the BAI round and r denotes a confidence radius; the two are never the same symbol. BAI maintains confidence intervals for active gradients and eliminates a generator i at round t if

$$|\hat{g}_i| + r_i < \max_{j \in \mathcal{A}_t} (|\hat{g}_j| - r_j), \quad (9)$$

where $\mathcal{A}_t$ is the active generator set and $r_i = z_\delta \epsilon_i$ is the current confidence radius of arm i , with ϵ_i its one-standard-error target and z_δ the confidence factor of Eq. (20). All methods in this report use a single common radius $r_i = r$ across active arms; the rule is written with per-arm radii because the covariance-aware variant of Sec. 14 does not. M2 does not exploit universal cross-gradient sharing.

Implementation status. M2 is now implemented as its own module. Each run writes the individual-gradient FC grouping $\mathcal{M}_{i,k}$ for every generator, the oracle fragment standard deviation of every fragment, the per-arm shot cost, and the full BAI elimination history with confidence radii and active sets. The M2 column is therefore auditable; it remains an oracle planning diagnostic rather than a finite-shot result.

4.3 M3: BAI-FC-UG

M3 combines the mechanisms of M1 and M2. It starts with the same fixed parent universal FC contexts as M1,

$$\mathcal{M}_1, \dots, \mathcal{M}_K, \quad (10)$$

but BAI eliminates generator candidates,

$$\mathcal{A}_0 \supset \mathcal{A}_1 \supset \mathcal{A}_2 \supset \dots. \quad (11)$$

At BAI round t , the active Pauli support is

$$\mathcal{B}_t = \bigcup_{i \in \mathcal{A}_t} \text{supp}(C_i), \quad \mathcal{B}_t \subseteq \mathcal{B}_0. \quad (12)$$

The active measurement contexts are restricted parent contexts,

$$\mathcal{M}_\alpha^{(t)} = \mathcal{M}_\alpha \cap \mathcal{B}_t. \quad (13)$$

The parent contexts are not rebuilt during BAI. This nested construction keeps previous measurements compatible while allowing the algorithm to stop sampling information that only affects eliminated generators.

How M3's cost is aggregated. Shots taken in a context are reusable, so a context is charged once, at the largest number of shots any round ever demanded of it. Writing $n_\alpha^{(t)}$ for the allocation of Eq. (21) solved over the active set $\mathcal{A}_t$ at the radius of round t , the reported M3 cost is

$$M_{\text{M3}} = \sum_{\alpha} \max_t n_\alpha^{(t)}. \quad (14)$$

Each round's allocation is solved independently, without reference to shots already committed; the maximum is taken afterwards. This is the accounting a real run would get for free by keeping its data, and it is why the trajectory is fully determined by the finitely many rounds at which the active set changes.

5 Oracle shot proxy

All numbers in Secs. 9–10 are oracle planning estimates: the planner is given the exact g_i and $\sigma_{i\alpha}$, and nothing is learned from shots. *Oracle* and *noiseless* mean only that. They do not refer to a noiseless device, and state-preparation error is excluded from the model altogether (Sec. 7).

Sorted order and the gap. Order the pool by decreasing absolute gradient, $|g_{(1)}| \geq |g_{(2)}| \geq \dots$; the parenthesised index is a rank, never a generator index. Then

$$\text{gap} = |g_{(1)}| - |g_{(2)}|, \quad \Delta_i = |g_{(1)}| - |g_i| \geq 0, \quad (15)$$

so $\Delta_{(2)} = \text{gap}$ and $\Delta_{(1)} = 0$.

Shots and context-shots. One *shot* is one state preparation plus one measurement of one context: a single circuit execution. *Context-shot* is the same thing charged to the context it was spent in; every cost quoted is $\sum_{\alpha} n_{\alpha}$. The metric counts shots only — not circuit depth, not the number of *distinct* measurement settings, not classical post-processing (Sec. 7).

For one gradient operator written as a sum of independently measured FC fragments,

$$C_i = \sum_{\alpha=1}^{K_i} F_{i\alpha}, \quad (16)$$

with oracle fragment variances

$$\sigma_{i\alpha}^2 = \text{Var}_{\psi}(F_{i\alpha}) = \langle F_{i\alpha}^2 \rangle_{\psi} - \langle F_{i\alpha} \rangle_{\psi}^2, \quad (17)$$

the optimal independent-fragment allocation gives the approximate total number of shots needed to estimate g_i with one-standard-error target ϵ_i as

$$M_i(\epsilon_i) \approx \frac{\left(\sum_{\alpha=1}^{K_i} \sigma_{i\alpha} \right)^2}{\epsilon_i^2}. \quad (18)$$

Precision target and confidence radius. ϵ_i is a one-standard-error target on $\hat{g}_i$; a *confidence radius* is the half-width of the corresponding interval,

$$r_i = z_{\delta} \epsilon_i, \quad (19)$$

so a method asked for radius r must reach $\epsilon = r/z_{\delta}$, and ϵ_i in Eq. (18) is replaced by r_i/z_{δ} . The confidence factor is the two-sided normal quantile with a Bonferroni correction over the whole pool,

$$z_{\delta} = \Phi^{-1} \left(1 - \frac{\delta}{2N} \right), \quad N = |\{G_i\}|, \quad (20)$$

with N the *full* pool size — 26, 92, 140 — not the non-zero count and not the active set, and no correction over rounds. At $\delta = 0.05$ this is $z_{\delta} = 3.10, 3.46$ and 3.57 , applied identically to all four constructions so that it cancels from every planning-bound ratio (Sec. 12). $\delta = 0.05$ is the conventional level; Bonferroni is chosen because it is the only family-wise correction needing no assumption about how the arms are correlated, which matters here because M1 and M3 read several arms off the same shots. The price is conservatism, and Sec. 11.5 measures it: the construction does not control the realised selection-error rate, so it is a convention rather than a guarantee.

Eq. (18) is applied separately inside each gradient arm, because M2 measures each arm in its own contexts. M1 and M3 instead share contexts between generators, so the allocation over contexts is not separable and is obtained by solving the convex programme

$$\min_{n_{\alpha} \geq 0} \sum_{\alpha} n_{\alpha} \quad \text{subject to} \quad \text{Var}(\hat{g}_i) = \sum_{\alpha} \frac{\sigma_{i\alpha}^2}{n_{\alpha}} \leq \epsilon^2 \quad \text{for every active } i, \quad (21)$$

where n_{α} is the number of shots taken in context α and $\text{Var}(\hat{g}_i) = \sum_{\alpha} \sigma_{i\alpha}^2 / n_{\alpha}$ is the *reconstruction variance* of gradient i under that allocation: the variance of the estimate obtained by reading $\hat{g}_i$ off the shared contexts through Eq. (5). The quantity $\sum_{\alpha} \sigma_{i\alpha}^2$ that appears repeatedly below is its value at one shot per context, and serves as a scale-free measure of how expensive generator i is to reconstruct from the parent grouping. Its stationarity condition is $n_{\alpha} = \sqrt{\sum_i \lambda_i \sigma_{i\alpha}^2}$ for multipliers $\lambda_i \geq 0$, solved by a damped multiplicative fixed-point iteration (damping 0.5, at most 500 iterations, stopping when the least-loaded constraint is within 10^{-9} of the tightest) and rescaled so the tightest constraint holds exactly. The rescaling makes the result feasible whatever the iteration converged to, so

the stopping rule affects optimality and never validity. For a single gradient it reduces exactly to Eq. (18), and it matches a general convex solver on the instances tested (Sec. 18).

Shot counts are not rounded. n_α is real-valued with no per-context minimum; the per-arm closed form of Eq. (27) is rounded up per arm. A floor of one shot per context would add at most K shots, at most 0.2% of any entry in Table 3 (worst case: 53 contexts under M3’s 33,432 on H_4 side 2.0 CISD), changing quoted digits but no ratio.

The schedule-free limit. As the common radius r decreases continuously, generator i leaves the active set the moment $2r$ falls below Δ_i , so it is resolved to exactly $\Delta_i/2$ and the leader survives to gap/2. Measurements are reusable, so an arm leaving at radius r has paid for radius r and no more, and a shared context has paid the largest requirement it ever saw. The active set changes only at the distinct values of Δ_i , so evaluating the allocation there determines the whole trajectory; for M2 it collapses to Eq. (27). This removes the free shrink factor of the previous revision, which being coarse overshoot every elimination point (Sec. 12).

Because these formulae depend on the exact grouping and fragment variances, every production run saves the FC groups, the fragment variances and the BAI histories used to generate each table entry.

6 Computational details

6.1 Molecular input

All cases use the minimal STO-3G basis, charge 0, and a closed-shell singlet restricted Hartree–Fock reference, with no frozen core and no qubit tapering. Water is placed in the xz plane with the oxygen at the origin and the bond angle held at 104.52° while $R(\text{OH})$ is varied, so the stretch is symmetric. The set is chosen to vary pool size at fixed basis and method — 26, 92 and 140 generators on 8, 12 and 14 qubits — while staying small enough that the exact FCI energy and exact fragment variances are available, since every number here depends on having them. The stretched geometries (LiH at 3.0 Å, H_4 at side 2.0 Å, H_2O at 1.75 Å) put each system in a strongly correlated regime where the leading gradients crowd together and selection is hard, and the two H_2O geometries isolate that effect at fixed pool size. Four systems cannot separate size from geometry and state, and no claim in this report about a trend with size should be read as established.

Geometries, in Å: LiH is Li at the origin and H at (0, 0, 3.0), 6 spatial orbitals. H_4 is a planar square in the xy plane, H at (0, 0, 0), (s , 0, 0), (0, s , 0), (s , s , 0) with $s = 1.0$ and 2.0, 4 orbitals. H_2O has O at the origin and H at $(\pm R \sin \frac{\theta}{2}, 0, R \cos \frac{\theta}{2})$ with $\theta = 104.52^\circ$ and $R = 0.9572$ and 1.75, 7 orbitals.

Two states are used per geometry. **HF** is the closed-shell Hartree–Fock determinant. **CISD** (configuration interaction with singles and doubles) is the lowest eigenvector of the qubit Hamiltonian restricted to the determinants within double excitation of that reference, in the target (N_α, N_β) sector, normalised to unity and embedded in the full 2^n -dimensional space. The CISD state is used as a correlated proxy for a later ADAPT state: it is a correlated wavefunction in the same sector, cheap to construct exactly, and it shifts the gradient spectrum in the direction a converging ADAPT run does — the top gradient falls by a factor of two to forty-four (Table 3), which is what makes selection expensive late in a trajectory. That is the whole of the justification. It is not a saved trajectory snapshot, and no evidence is offered that its gradient *spectrum* resembles one; testing that needs trajectory data and is listed in Sec. 17. LiH is run with the HF reference only.

6.2 Generator pool, conventions and labels

Spin orbitals are interleaved: spatial orbital p gives spin orbitals $2p$ (α) and $2p + 1$ (β), zero-based. The computational basis is indexed big-endian, matching the sparse-operator convention of the linear-algebra backend. The Jordan–Wigner transform is used throughout.

The generator universe is the standard spin-orbital UCCSD pool. Each generator is anti-Hermitian,

$$G_i = \mathcal{JW}(\tau_i - \tau_i^\dagger), \quad \tau_i^S = a_a^\dagger a_i, \quad \tau_i^D = a_a^\dagger a_b^\dagger a_j a_i, \quad (22)$$

so that $g_i = \langle [H, G_i] \rangle_\psi$ of Eq. (1) is real and equals $\partial E / \partial \theta_i$ at $\theta_i = 0$ for the parameterisation $e^{\theta_i G_i}$. Singles run over occupied–virtual pairs of the same spin; doubles conserve N_α and N_β separately. The pool therefore conserves electron number and S_z but not generally S^2 . Ordering is deterministic: all singles in ascending (i, a) , then all doubles in ascending (i, j, a, b) . The labels used in the tables are exactly this: S $i \rightarrow a$ for a single and D $i, j \rightarrow a, b$ for a double, with zero-based interleaved spin-orbital indices. The resulting pool sizes are 26, 92 and 140 for H_4 , LiH and H_2O . This is the pool of the original ADAPT-VQE construction [1], chosen so that the observable structure in Sec. 8 is the standard one; qubit and spin-adapted pools give different supports and different pool sizes, and the comparison has not been repeated for them.

6.3 Fully commuting grouping routine

FC group counts are routine-dependent, and two rows are comparable only if produced by the same deterministic routine with the same Pauli ordering, tie-breaking rule and seed. Every row in the main tables uses one routine: deterministic greedy first-fit grouping, with Pauli strings sorted by decreasing weight (the number of non-identity factors) and ties broken by ascending lexicographic order of the string read left to right from qubit 0, under the letter order $I < X < Y < Z$, with each candidate joining the lowest-numbered existing group that accepts it and opening a new group otherwise. No random seed is used. Sec. 13 varies this convention deliberately and reports the spread. The counts differ from the previous revision, whose H_4 and H_2O rows used a different partitioning routine and were not comparable with the LiH row.

Fully commuting grouping is chosen over qubit-wise commuting because it admits strictly more terms per context and so minimises the quantity this report counts. The cost of that choice is a Clifford diagonalisation circuit per context, which the shot metric does not price (Sec. 7); a qubit-wise-commuting comparison would trade more contexts for cheaper circuits and is the natural companion study once circuits are priced.

6.4 Validation gate

Every case is checked before any gradient is computed: the Jordan–Wigner (JW) Hamiltonian must reproduce the PySCF restricted Hartree–Fock (RHF) energy on the Hartree–Fock (HF) determinant and the PySCF full configuration interaction (FCI) energy as its lowest eigenvalue in the (N_α, N_β) sector, both to 10^{-8} Ha. All nine cases pass, with a maximum observed error of 4×10^{-15} Ha. A case that fails the gate aborts rather than producing numbers.

6.5 Non-zero gradient threshold

Gradients that vanish by symmetry appear at the 10^{-8} numerical noise floor of the integral and SCF pipeline, while the smallest physically non-zero gradient in these cases is of order 10^{-3} . The counts below use a threshold of 10^{-6} , which lies on the plateau between the two. The threshold is not innocuous at tighter values: for LiH the count is 26 at 10^{-6} and 34 at 10^{-9} , and every run records the count at 10^{-4} , 10^{-6} , 10^{-8} and 10^{-9} so the choice stays visible. **The threshold is used for reporting only.** No method filters the pool by it: all four constructions carry every one of the N generators as an arm, and N in Eq. (20) is the full pool size. The one place the distinction is used analytically is the diagnostic in Sec. 9.1, which asks what M1 would cost if it were given the non-zero set for free.

7 What the model assumes and excludes

Statistical assumptions, in one place. (i) Estimators are Gaussian; the normal limit is assumed, not tested against shot-level sampling of measurement outcomes. (ii) Distinct measurement contexts are sampled independently. (iii) Within a context, the planning bounds and the main finite-shot tables treat different generators’ fragments as independent; this is exact for M2 and the baseline, which give every generator its own shots, and an approximation for M1 and M3. Sec. 14 removes it and reports what changes. (iv) State preparation is exact and noiseless, and there is no gate, readout or decoherence error anywhere in the model. (v) The union bound behind Eq. (20) is taken over the N pool members only; it is not taken over the sequence of elimination rounds, which in the exact limit is a continuum. (vi) Fragment variances are the exact $\text{Var}_\psi(F_{i\alpha})$, so no cost is charged for learning them. (vii) In the finite-shot surrogate each round’s estimate is drawn afresh at the accumulated shot count rather than as an increment on the previous draw, so successive rounds are treated as independent where real data would be strongly correlated. This is a simplification, not a conservative bound in either direction, and its effect on the error rates and run lengths has not been measured.

“System size” means the pool size N , ordering H_4 (26) < LiH (92) < H_2O (140). On these systems that agrees with qubit count (8, 12, 14) and $|\mathcal{B}_0|$ (1,276; 30,196; 81,566) but not with electron count. With four systems, and geometry and state varying with them, no claim here about a trend with size is established; the phrase labels the ordering.

What the cost metric leaves out. Circuit depth and two-qubit count for the diagonalising Cliffords, compilation and calibration of distinct measurement settings, and classical post-processing. The number of distinct settings differs sharply between the methods, against the individually grouped ones.

M2 and the baseline need 3.5 to 6.1 times as many distinct settings as M1 and M3 here. Nothing in this report prices that, and a metric that did could move the M2-versus-M3 comparison, which on shots alone is already within a factor of two on most rows. The counts are structural properties of the groupings, not measured circuit costs.

Table 1: Distinct measurement settings each construction requires, a cost component the shot metric does not price. M1 and M3 share one parent grouping; M2 and the baseline group every generator separately, so their settings are the sum over arms and are not reused between arms.

Case	M1, M3: K	M2, baseline: $\sum_i K_i$	Largest single arm: $\max_i K_i$
LiH, $R = 3.0 \text{ \AA}$	980	5,744	101
H ₄ square, side 1.0 and 2.0 $\text{\AA}$	53	183	10
H ₂ O eq., $R(\text{OH}) = 0.9572 \text{ \AA}$	2,366	13,235	159
H ₂ O stretch, $R(\text{OH}) = 1.75 \text{ \AA}$	2,356	14,263	162

Table 2: Global universal support used by M1: FC-UG and M3: BAI-FC-UG. The FC contexts are global parent contexts for $\mathcal{B}_0$, not per-gradient groupings, and all rows use the single greedy routine fixed in Sec. 6. Non-zero gradient counts use the 10^{-6} threshold. Distances in $\text{\AA}$.

Case	State	Qubits	Pool	Universal terms	FC-UG contexts	Nonzero g_i
LiH, $R=3.0$	HF	12	92	30,196	980	26
H ₄ , side 1.0	HF	8	26	1,276	53	10
H ₄ , side 1.0	CISD	8	26	1,276	53	10
H ₄ , side 2.0	HF	8	26	1,276	53	10
H ₄ , side 2.0	CISD	8	26	1,276	53	10
H ₂ O eq., $R=0.9572$	HF	14	140	81,566	2,366	40
H ₂ O eq., $R=0.9572$	CISD	14	140	81,566	2,366	48
H ₂ O str., $R=1.75$	HF	14	140	81,566	2,356	40
H ₂ O str., $R=1.75$	CISD	14	140	81,566	2,356	48

8 Observable structure for all systems

The universal supports reproduce the earlier pilot data exactly (30,196 for LiH, 1,276 for H₄, 81,566 for H₂O), as do the gradient values and gaps of Table 3 to six or more digits. The FC context counts differ from the pilot’s because they now all come from one routine.

Why the two H₂O geometries have the same support size but different context counts. The grouping routine is deterministic in the set of Pauli strings alone, so equal supports would have to give equal counts; 2,366 against 2,356 therefore means the supports are not equal. They are not. Both contain 81,566 products, and $\sum_i |\text{supp}(C_i)|$ is 314,720 in both, but their intersection is only 78,113: each geometry has 3,453 Pauli products the other lacks, a symmetric difference of 6,906, or 8.5% of each. The differing products are not numerical dust — their largest commutator coefficients reach 0.09 and 0.11, and their medians, 2.6×10^{-3} and 3.7×10^{-3} , are within a factor of two of the median over the shared products.

The cause is an orbital reordering. Both geometries have the same point group and the same multiset of orbital symmetries, but the orbital energies cross: the canonical order is $(A_1, A_1, B_2, A_1, B_1, A_1, B_2)$ at 0.9572 $\text{\AA}$ and $(A_1, A_1, B_1, B_2, A_1, A_1, B_2)$ at 1.75. The symmetry-allowed integrals are equal in number, which is why $|\mathcal{B}_0|$ and $\sum_i |\text{supp}(C_i)|$ agree exactly, but they sit at different orbital indices and so map to different Pauli strings. The same reordering moves the top generator from $\mathcal{D} \ 4,5 \rightarrow 12,13$ to $\mathcal{D} \ 6,7 \rightarrow 12,13$. The two geometries are comparable as physics, their Pauli-level structures are not identical, and a difference of ten contexts carries no information.

9 The complete comparison

All four constructions are compared at one criterion: the winner must be separated from the rest of the pool at family-wise confidence $\delta = 0.05$. M1 and the baseline reach that with a single common radius gap/2 over the whole pool; M2 and M3 reach it by elimination, resolving each generator only to $\Delta_i/2$. No free schedule or precision parameter remains in any column. The cost scales as the inverse square of the relevant deficit, so the gap column is the main driver of difficulty.

M1 is given the exact gap, and this favours it. A non-adaptive method cannot know gap before measuring anything, yet the target radius gap/2 is computed from the exact gradients. M1 is therefore charged the loosest radius that happens to suffice, the most generous reading available; On H₄ side 1.0 HF the M1/M2 ratio moves from 0.66 to 168 as the target radius gap/ d varies over $d = 1, 2, 4, 8, 16$, M1 costing 561,216, 2,244,862, 8,979,447, 35,917,787 and 143,671,148 shots. The same oracle knowledge sets the good-enough tolerance of Sec. 11.6. Read every M1 column as *M1 at its best*.

The winners are $\mathcal{D} \ 2,3 \rightarrow 10,11$ (LiH), $\mathcal{D} \ 2,3 \rightarrow 6,7$ (both H₄ HF rows), $\mathcal{D} \ 0,1 \rightarrow 6,7$ (both H₄ CISD rows), $\mathcal{D} \ 4,5 \rightarrow 12,13$ (H₂O equilibrium) and $\mathcal{D} \ 6,7 \rightarrow 12,13$ (H₂O stretched), with $|g_{(1)}|$ of 0.243187, 0.295316, 0.142467, 0.315036, 0.098961, 0.304910, 0.006925, 0.388493 and 0.106873 down the rows. All four constructions return the true maximiser in all nine cases, which in oracle mode is a consistency check rather than a result: the rule reads exact gradients, so it cannot eliminate the maximiser. Whether it is still found once the gradients are estimated is the subject of Sec. 11.

Table 3: Oracle context-shot comparison at the winner-identification criterion, $\delta = 0.05$. $M1/M2 > 1$ means M2 is cheaper than M1; $M2/M3 > 1$ means M3 is cheaper than M2. Entries below unity are bold. Distances in Å; the baseline is the no-sharing diagnostic of Sec. 10.

Case	State	Gap	Baseline	M1	M2	M3	M1/M2	M2/M3	M1/M3
LiH, $R=3.0$	HF	0.096388	6,819,980	3,563,904	1,140,583	638,423	3.12	1.79	5.58
H ₄ , side 1.0	HF	0.021539	4,848,846	2,244,862	854,629	566,951	2.63	1.51	3.96
H ₄ , side 1.0	CISD	0.118081	130,375	59,642	102,806	53,727	0.58	1.91	1.11
H ₄ , side 2.0	HF	0.016649	2,002,941	598,102	422,746	220,389	1.41	1.92	2.71
H ₄ , side 2.0	CISD	0.051859	180,604	58,068	86,857	33,432	0.67	2.60	1.74
H ₂ O eq., $R=0.9572$	HF	0.101090	204,022,319	172,147,028	25,096,606	22,130,852	6.86	1.13	7.78
H ₂ O eq., $R=0.9572$	CISD	0.002064	485,001,056,522	411,524,732,152	46,457,815,707	44,971,732,317	8.86	1.03	9.15
H ₂ O str., $R=1.75$	HF	0.021856	4,407,619,149	3,864,803,902	17,423,631	20,163,542	221.81	0.86	191.67
H ₂ O str., $R=1.75$	CISD	0.016957	6,317,143,181	5,715,348,735	168,305,383	165,843,674	33.96	1.01	34.46

M3 is cheaper than M1 in every row. The intuition is that M3 solves the same allocation over a shrinking subset of the constraints at radii never tighter than M1’s; but the reported M3 cost is the sum over contexts of the per-context maximum across rounds, Eq. (14), and a sum of maxima of separately optimised allocations is not obviously bounded by the optimum of the full problem. **No proof is offered.** The ordering is checked numerically on every case and holds; read it as an observation with a plausible mechanism. M3 is cheaper than M2 in eight rows of nine and M2 than M1 in seven, both subject to the routine-dependence of Sec. 13.

9.1 What sets the M1 cost

M1 must reach its common radius for every generator simultaneously, so its cost is set by the pool member with the largest reconstruction variance $\sum_{\alpha} \sigma_{i\alpha}^2$ — the *binding generator*, in the sense that its constraint in Eq. (21) is the one the final rescaling makes tight. Table 4 identifies it in every case.

Table 4: The generator whose constraint binds M1, against the winner. $\sum_{\alpha} \sigma^2$ is the reconstruction variance at one shot per parent context. The last column is the factor by which M1’s cost falls if it is handed the set of non-zero-gradient generators for free — information no non-adaptive method has.

Case	Binding generator	$ g $ of it	$\sum_{\alpha} \sigma^2$	same, winner	M1 all / M1 non-zero
LiH, $R=3.0$, HF	D 0,1 $\rightarrow$ 4,9	0	11.76	0.072	2.28
H ₄ , side 1.0, HF	D 0,1 $\rightarrow$ 4,7	0	2.713	0.447	1.71
H ₄ , side 1.0, CISD	D 0,1 $\rightarrow$ 6,7	1.4×10^{-1}	1.741	1.741	1.56
H ₄ , side 2.0, HF	D 0,1 $\rightarrow$ 4,7	0	0.218	0.067	1.83
H ₄ , side 2.0, CISD	S 1 $\rightarrow$ 7	1.9×10^{-31}	0.144	0.122	1.93
H ₂ O eq., HF	S 0 $\rightarrow$ 10	7.1×10^{-14}	537.4	1.317	2.20
H ₂ O eq., CISD	S 0 $\rightarrow$ 10	1.6×10^{-5}	530.8	1.284	2.01
H ₂ O str., HF	S 0 $\rightarrow$ 10	1.8×10^{-8}	531.6	0.592	2.35
H ₂ O str., CISD	S 0 $\rightarrow$ 10	2.0×10^{-4}	455.9	0.565	2.02

In six of the nine cases the binding generator has a gradient below the 10^{-6} threshold, and its reconstruction variance dwarfs the winner’s — by a factor of 164 on LiH and of 400 to 900 on H₂O. In the remaining three it carries a small but non-zero gradient, and on H₄ side 1.0 with the CISD state it is the winner itself.

This is not a defect of the comparison: M1 is non-adaptive, so it must bound every generator tightly enough to rule it out, and paying for high-variance operators that turn out not to matter is precisely the inefficiency elimination removes. Nor is it the whole story. Restricting M1 to the non-zero-gradient generators — the last column of Table 4 — cuts its cost by only 1.6 to 2.4, because the rest also carry substantial variance. The cost is spread over the pool rather than concentrated in one irrelevant operator. This is also a rough guide to how much Part III symmetry filtering could do for M1, although not a bound: filtering and tapering transform the operators as well as dropping generators, so the two are not the same operation.

9.2 Where the cost actually sits: breadth and depth

The shot cost of a BAI method is not concentrated in the final elimination, as a naive reading of the $1/r^2$ scaling suggests. It splits into two terms, and which one dominates changes with the system.

One definition, applied the same way to both methods. Let $r_0 = \frac{1}{2} \max_i \Delta_i$ be the coarsest radius of the trajectory, the radius at which the first generator is eliminated and at which every generator is still active. *Breadth* is the cost of that first sweep,

$$M^{\text{breadth}} = \text{cost of resolving every generator to } r_0, \quad M^{\text{depth}} = M^{\text{total}} - M^{\text{breadth}}, \quad (23)$$

and the *breadth share* is $M^{\text{breadth}}/M^{\text{total}}$. For M2 the cost is separable, so $M^{\text{breadth}} = \sum_i (z_{\delta} S_i^{\text{own}}/r_0)^2$, writing $S_i^{\text{own}} = \sum_{\alpha} \sigma_{i\alpha}$ over generator i ’s own contexts and S_i^{par} for the same sum over the parent contexts. For M3 it is the total of the allocation at the first breakpoint, before any elimination. Because shots are reusable, breadth is a cost neither method can avoid: it is the price of learning which generators are worth pursuing.

The two HF H₄ rows are strongly depth-dominated: only 1 to 3% of the cost is committed before the first elimination, and essentially all of it goes into separating the leading pair. LiH, both H₂O geometries and the two

Table 5: Breadth share, Eq. (23): the fraction of each method’s total that is already committed at the coarsest radius, when every generator is still active. The remainder is the depth term. Both columns use the same definition; M3’s is the cumulative allocation at the first breakpoint of its elimination history.

Case	M2 breadth share	M3 breadth share	$\max_i \Delta_i/\text{gap}$
H ₄ , side 1.0, HF	3.0%	2.1%	13.7
H ₄ , side 1.0, CISD	87.1%	76.3%	1.2
H ₄ , side 2.0, HF	1.3%	0.8%	18.9
H ₄ , side 2.0, CISD	57.1%	47.7%	1.9
LiH, $R=3.0$, HF	93.9%	87.7%	2.5
H ₂ O eq., HF	89.4%	85.5%	3.0
H ₂ O eq., CISD	92.8%	81.3%	3.4
H ₂ O str., HF	80.1%	60.7%	17.8
H ₂ O str., CISD	94.5%	86.8%	6.3

H₄ CISD rows are breadth-dominated, with 57 to 94% of the cost already spent when every generator is still in play. What distinguishes them is $\max_i \Delta_i/\text{gap}$, the dynamic range the schedule has to cover: 13.7 and 18.9 on the two depth-dominated H₄ HF rows against 1.2 to 3.4 on H₄ CISD, LiH and H₂O equilibrium. Where the leading pair is nearly tied relative to the spread of the field, depth dominates; where the deficits are spread more evenly, breadth does. H₂O stretch, at 17.8 and 6.3, does not fit that reading, so the correlation is suggestive rather than established.

The two terms reward sharing very differently. In the breadth term a shared context serves the entire pool, 92 or 140 generators at once, and M3’s advantage over M2 is correspondingly large. In the depth term it serves only the generators that survive to the final radius. M3’s breadth share is below M2’s in every one of the nine rows, which is what sharing is supposed to do: it makes the coarse sweep cheaper relative to the rest.

9.3 What decides the depth term

Two quantities decide it. Generator i ’s own FC grouping is $\mathcal{M}_{i,1}, \dots, \mathcal{M}_{i,K_i}$; write $F_{i\alpha}$ for the part of C_i landing in parent context $\mathcal{M}_\alpha$. The *fragmentation penalty*

$$f_i = \left(\frac{\sum_\alpha \sigma_{i\alpha} \text{ over parent contexts}}{\sum_k \sigma_{ik} \text{ over its own } \mathcal{M}_{i,k}} \right)^2 \quad (24)$$

is the factor by which the shared grouping makes one gradient more expensive to resolve alone. It exceeds unity in every case here because Eq. (18) depends on $(\sum_\alpha \sigma_{i\alpha})^2$ rather than $\sum_\alpha \sigma_{i\alpha}^2$, so finer splitting raises the sum of standard deviations — but this is an *observation*, not a theorem: the parent grouping is not a refinement of the own grouping, since a parent context can merge Paulis the own grouping separates, so a larger context count does not by itself imply $f_i \geq 1$. The second quantity is $|\mathcal{A}| = |\{i : \Delta_i \leq \text{gap}\}|$, the number of generators still active at the final radius; by construction $|\mathcal{A}| \geq 2$.

In Table 6, K^{own} counts groups in the winner’s own FC grouping and depends only on $\text{supp}(C_i)$, not the state; K^{parent} counts parent contexts in which the winner has a *non-zero* fragment standard deviation, which is state-dependent — hence 858 against 887, and 875 against 899, on the H₂O HF and CISD rows for the same winner. Contexts with $\sigma_{i\alpha} = 0$ contribute nothing and are not counted.

Table 6: Fragmentation of the winning gradient against the number of finalists. K^{own} is the number of groups in the winner’s own FC grouping; K^{parent} is the number of parent contexts in which it has a non-zero fragment standard deviation, which is why it depends on the state. Both are defined in the text above.

Case	State	K^{own}	K^{parent}	f	$ \mathcal{A} $	M2/M3
H ₄ , side 1.0	HF	10	22	1.13	2	1.51
H ₄ , side 1.0	CISD	7	28	1.07	3	1.91
H ₄ , side 2.0	HF	10	22	1.19	3	1.92
H ₄ , side 2.0	CISD	7	28	1.22	2	2.60
LiH, $R=3.0$	HF	58	520	3.49	3	1.79
H ₂ O eq., $R=0.9572$	HF	57	858	2.54	2	1.13
H ₂ O eq., $R=0.9572$	CISD	57	887	2.44	2	1.03
H ₂ O str., $R=1.75$	HF	50	875	3.62	2	0.86
H ₂ O str., $R=1.75$	CISD	50	899	2.94	2	1.01

The scaling looks asymmetric. $|\mathcal{A}|$ is set by how many generators are nearly tied at the top, which has nothing to do with pool size and is two or three in every case here. The fragmentation penalty is larger on the larger systems, because a larger universal support spreads each individual gradient over more parent contexts: 22 on H₄, 520 on LiH, roughly 875 on H₂O. If that persisted, M3’s benefit in the depth term would have a ceiling while its cost in that term did not, and a crossover would follow.

This is a hypothesis, not a demonstration. f is not monotone in system size — LiH, the smaller system, gives 3.49 against 2.54 for H₂O at equilibrium — four systems cannot separate size from geometry and state, and f

grows far more slowly than the raw context count (3.6 against 17.5 on H₂O stretch), because most parent contexts hold a negligible sliver of any one gradient. The mechanism is real but weak, and nothing here fixes where or whether a crossover occurs. Sec. 11 finds the opposite trend under finite shots; Sec. 11.4 reconciles them.

9.4 The finalists do not share Pauli terms

The premise of universal grouping is that gradient operators share Pauli products. Across the pool they do: each Pauli product in $\mathcal{B}_0$ appears in roughly four different gradient commutators.

Table 7: Support sharing. Average reuse is $\sum_i |\text{supp}(C_i)|/|\mathcal{B}_0|$. Finalist overlap is the Jaccard overlap $|\text{supp}(C_{(1)}) \cap \text{supp}(C_{(2)})|/|\text{supp}(C_{(1)}) \cup \text{supp}(C_{(2)})|$ between the supports of the two leading generators. Supports do not depend on the state, but the identity of the finalists does, so the two H₄ rows differ. The remaining four cases have not been evaluated.

Case	$ \mathcal{B}_0 $	$\sum_i \text{supp}(C_i) $	Average reuse	Finalist overlap	M2/M3
H ₄ , side 1.0, HF	1,276	4,148	3.25	0.0%	1.51
H ₄ , side 1.0, CISD	1,276	4,148	3.25	27.8%	1.91
LiH, $R=3.0$, HF	30,196	120,384	3.99	35.1%	1.79
H ₂ O eq., $R=0.9572$, HF	81,566	314,720	3.86	0.0%	1.13
H ₂ O str., $R=1.75$, HF	81,566	314,720	3.86	0.0%	0.86

The two columns tell opposite stories. Pool-wide reuse of about four is why M3’s breadth term is efficient. But in three of the five cases the two finalists share *no* Pauli products at all, so at the depth stage there is nothing to share and M3 pays the fragmentation penalty for nothing.

The overlap column does not order the M2/M3 column across systems. The two cases with substantial finalist overlap, LiH and H₄ CISD, are cases where M3 keeps a clear advantage, and the H₂O rows with none are where it disappears — but H₄ HF has zero overlap and still gives 1.51, well above both H₂O rows, and its fragmentation penalty is six times smaller than LiH’s. Low finalist overlap is a plausible reason for M3 to lose its depth-stage advantage and is present where the advantage is smallest; five rows are not enough to make it *the* explanation.

9.5 An alternative: rebuild the grouping once the field narrows

If the fixed parent contexts are the problem at the depth stage, an obvious remedy is to abandon them once the active set is small and build a fresh fully commuting grouping over only the survivors’ union support, accepting that the earlier shots were taken in different circuit settings and do not transfer. This was tested directly. Phase one uses the parent contexts while more than a chosen number of generators remain active; phase two regroups from scratch, using the same greedy first-fit routine and ordering as the parent grouping, and is charged the full cost of reaching the final radii, ignoring the head start phase one has already provided, so the totals below are upper bounds. Switch points $|\mathcal{A}| \leq 2, 3, 4, 6, 10, 20$ were tested on each case and the best is reported; because the active set jumps in large steps, several thresholds often fire at the same breakpoint and give identical totals.

Table 8: Staged regrouping against the better of M2 and M3. The switch point is the active-set size at which the parent contexts are abandoned.

Case	Better of M2, M3	Best staged total	Switch at	Ratio
H ₄ , side 1.0, HF	566,951	502,858	$ \mathcal{A} \leq 10$	0.89
LiH, $R=3.0$, HF	638,423	638,423	never	1.00
H ₂ O str., $R=1.75$, HF	17,423,631	16,880,737	$ \mathcal{A} \leq 3$	0.97

Staged regrouping does beat both methods, but only by 3 to 11%, and the best switch point is not consistent: early on H₄, late on H₂O, and never on LiH, where the procedure degenerates to M3. The reason is visible in Table 5. Regrouping can only improve the depth term, and on the systems where M3 struggles the cost is dominated by breadth, which the remedy does not touch. On H₂O stretch it improves M3 by 16% and the better of the two methods by 3%.

A second observation bears on *where* regrouping should act. On H₄ side 1.0 HF the best switch is the earliest available: the first breakpoint, where sixteen of twenty-six generators leave at once and regrouping the ten survivors needs twelve contexts against fifty-three. The union support collapses disproportionately because the sixteen zero-gradient generators hold 708 of the 1,276 Pauli products in $\mathcal{B}_0$ (55.5%) and their supports are *disjoint* from the ten non-zero ones. This is one case, and the disjointness is a property of the H₄ square not measured elsewhere; subject to that, the leverage is not at the final pair but as soon as the first substantial group is ruled out — which is also where Part III symmetry filtering would act, without measuring anything first.

10 Attribution: the no-sharing baseline

The three methods are two binary choices, sharing and elimination, and they span only three of the four combinations. The fourth is a baseline that does neither: every generator is measured in its own fully commuting grouping,

and every one of them is resolved to the same common radius used by M1. It is reported here as a diagnostic outside the method hierarchy, and it serves two purposes. It makes the two mechanisms separately attributable, and it is a sanity bound, since measuring every generator separately is always available as a fallback.

Table 9: Improvement each mechanism buys against the no-sharing baseline of Table 3. Sharing only is baseline/M1, elimination only baseline/M2, both baseline/M3. The product column is what “both” would be if the mechanisms acted independently; the interaction is the ratio of the two.

Case	State	Sharing only	Elimination only	Both	Product	Interaction
LiH, $R=3.0$	HF	1.91	5.98	10.68	11.44	0.93
H ₄ , side 1.0	HF	2.16	5.67	8.55	12.25	0.70
H ₄ , side 1.0	CISD	2.19	1.27	2.43	2.77	0.88
H ₄ , side 2.0	HF	3.35	4.74	9.09	15.87	0.57
H ₄ , side 2.0	CISD	3.11	2.08	5.40	6.47	0.84
H ₂ O eq., $R=0.9572$	HF	1.19	8.13	9.22	9.63	0.96
H ₂ O eq., $R=0.9572$	CISD	1.18	10.44	10.78	12.30	0.88
H ₂ O str., $R=1.75$	HF	1.14	252.97	218.59	288.50	0.76
H ₂ O str., $R=1.75$	CISD	1.11	37.53	38.09	41.49	0.92

Two things follow, and both bear on how the project should be described.

The mechanisms are substitutes, not complements. The interaction is below unity in every one of the nine cases, between 0.57 and 0.96. Applying both mechanisms buys less than the product of applying each alone. This is the quantitative form of the argument in Sec. 9.2: elimination removes precisely the situation that sharing exploits, namely many generators active at once, so the second mechanism applied always finds less left to gain.

Sharing contributes less than elimination, and less on the larger systems. On its own, universal grouping is worth a factor of 2.2 to 3.4 on H₄, 1.9 on LiH, and only 1.1 to 1.2 on H₂O. Elimination on its own is worth between 1.27 and 252.97: the low end is the H₄ side 1.0 CISD row, where the gap is wide and there is little for elimination to save, and the high end is the H₂O stretch HF row. The two ranges overlap, so “sharing contributes little” should be read as *relative to elimination* and not as an absolute statement — a factor of 3.4 is a real saving on its own terms. What the table does establish is the ordering of the two mechanisms and its direction: the sharing factor is smallest on the largest system in this set, on all four H₂O rows. Whether that is a size effect or a property of these particular molecules cannot be settled from four systems.

11 Finite-shot behaviour

Every number above is an oracle planning bound. The elimination rule reads exact gradients, so it cannot misfire, and the family-wise confidence is asserted rather than measured. This section replaces the exact gradients with estimates built from a finite number of shots and runs the same elimination rule on those.

Protocol. A real run does not know the deficits in advance and cannot jump to the exact breakpoints, so each trial follows a geometric radius schedule: the common radius starts at $\max_i |g_i|$, wide enough that no arm is eliminated for free, and is multiplied by $\gamma = 0.5$ each round. At each round the allocation of Eq. (21) is solved for the current active set and radius, and each context’s shot count is raised to that value if it is higher, never lowered; the estimate $\hat{g}_i$ is then redrawn from $\mathcal{N}(g_i, \sum_{\alpha} \sigma_{i\alpha}^2/n_{\alpha})$ at the accumulated n_{α} , which is the distribution of the estimator built from all shots taken so far. The elimination rule of Eq. (9) is then applied. A trial ends when one arm remains or the radius falls below $10^{-3} \max_i |g_i|$. Every case uses 200 trials at seed 0; the trial count is set so that the standard error on a mean cost is a few per cent, which is the resolution the comparisons need. The starting radius and the floor are both taken from the exact $\max_i |g_i|$ and are therefore oracle quantities, though mild ones: a real run would take the starting radius from a cheap pilot estimate of the largest gradient, and the floor only prevents a non-terminating run. This is a weaker use of oracle knowledge than M1’s exact gap (Sec. 9) but not none, and it qualifies the statement there that the best-arm methods need no such input.

Most of the inflation is the schedule, not noise. The same coarse $\gamma = 0.5$ overpays by 1.64 to 3.80 at the planning level with no noise at all (Table 13), against realised inflations of 1.40 to 3.16. The two cannot be subtracted — the noiseless run resolves the leader to gap/2 while a trial stops when one arm remains — but discretisation clearly dominates, and a tuned schedule would recover most of the gap to the bound. What the schedule does *not* explain is the difference between the methods: at $\gamma = 0.5$ and no noise the M2/M3 ratio moves by only -4% to $+20\%$, against up to $+125\%$ realised (LiH, 1.79 to 4.03), so the differential of Sec. 11.3 is not a schedule artefact. The two deliberate approximations, the normal limit and independence between generators, are listed in Sec. 7; Sec. 14 removes the second.

Columns. *Bound* is the oracle planning cost of Table 3; *Infl.* is the mean total shots over the 200 trials divided by it; *Correct* is the fraction of trials returning the true maximum-gradient generator. M2/M3 columns are ratios of the corresponding entries, standard errors combined in quadrature.

Table 10: Realised cost against the planning bound, 200 trials per row at seed 0, all uncertainties one standard error of the mean. Every row selected the true maximiser in all 200 trials, a Wilson 95% interval of $[0.981, 1]$; the *Correct* column is therefore constant and omitted. The non-adaptive baseline and M1 realise their bound exactly, having no schedule to overshoot, and are omitted.

Case	State	M2: BAI-FC-IG		M3: BAI-FC-UG		M2/M3	
		Bound	Infl.	Bound	Infl.	Bound	Realised
LiH, $R=3.0$	HF	1,140,583	3.16 ± 0.06	638,423	1.40 ± 0.03	1.79	4.03 ± 0.12
H ₄ , side 1.0	HF	854,629	2.21 ± 0.08	566,951	2.33 ± 0.10	1.51	1.43 ± 0.08
H ₄ , side 1.0	CISD	102,806	2.99 ± 0.05	53,727	2.74 ± 0.04	1.91	2.09 ± 0.04
H ₄ , side 2.0	HF	422,746	2.33 ± 0.05	220,389	2.48 ± 0.05	1.92	1.81 ± 0.05
H ₄ , side 2.0	CISD	86,857	2.77 ± 0.06	33,432	2.95 ± 0.09	2.60	2.43 ± 0.09
H ₂ O eq., $R=0.9572$	HF	25,096,606	3.02 ± 0.06	22,130,852	1.53 ± 0.04	1.13	2.24 ± 0.07
H ₂ O str., $R=1.75$	HF	17,423,631	2.75 ± 0.06	20,163,542	1.66 ± 0.04	0.86	1.43 ± 0.04

Realised means are Bound $\times$ Infl.; the largest is 7.6×10^7 shots for M2 on H₂O at equilibrium and the smallest 9.9×10^4 for M3 on H₄ side 2.0 CISD.

The two H₂O CISD rows are absent because they have not been simulated; they are the most expensive cases and were left out of this round rather than excluded for any reason bearing on the result (Sec. 17).

The scatter is not negligible and is the reason the errors are quoted. Repeated runs of the same configuration at different seeds move a realised total by 2 to 4%, which is comparable to some of the differences between adjacent rows and larger than the difference between the bound and realised M2/M3 ratios on the H₄ HF rows.

11.1 Finite shots reorder the comparisons with M1

M1 and the no-sharing baseline are non-adaptive: they solve one allocation at one radius and realise their planning bound exactly, with no schedule to overshoot. The best-arm methods do not. Comparing realised cost against M1’s bound therefore gives a different ranking from Table 3, and on several rows it reverses.

Table 11: M1 against the *realised* cost of the best-arm methods. M1’s bound is its realised cost. Entries below unity, where the best-arm method is more expensive than static all-gradient estimation, are set in bold.

Case	State	M1	M1 / M2 realised	M1 / M3 realised
LiH, $R=3.0$	HF	3,563,904	0.99	3.99
H ₄ , side 1.0	HF	2,244,862	1.19	1.70
H ₄ , side 1.0	CISD	59,642	0.19	0.40
H ₄ , side 2.0	HF	598,102	0.61	1.09
H ₄ , side 2.0	CISD	58,068	0.24	0.59
H ₂ O eq., $R=0.9572$	HF	172,147,028	2.27	5.07
H ₂ O str., $R=1.75$	HF	3,864,803,902	80.7	115.3

On the planning bounds M3 beats M1 on all seven of these rows and M2 on five. On realised cost M3 beats M1 on five and M2 on three. Both H₄ CISD rows invert for M3, which won them on the bounds; M2 had already lost them at bound level and now loses two more, and LiH becomes a tie for M2. The pattern is that elimination pays where M1 is expensive — H₂O, where the factor stays between 2 and 115 — and loses where M1 is already cheap, which is exactly the regime the planning bounds identified in conclusion 4, but the finite-shot penalty moves two more rows across the line.

Some of this is the coarse schedule rather than noise, as noted above, and a deployment would tune the schedule. It nonetheless means the headline factors of 1.1 to 192 from Table 3 are the *bound-level* advantage of M3 over M1, and that the realised advantage is smaller everywhere and negative on two of seven rows.

11.2 The one inversion does not survive

On H₂O stretch with the Hartree–Fock reference, the planning bounds give $M2/M3 = 0.86$, the only row in which M3 is more expensive than M2. Under finite shots the same row gives 1.43 ± 0.04 , and M3 wins. The inversion is an artefact of comparing two bounds that inflate by different amounts, and the conclusion drawn from it in the previous revision is retracted in Sec. 15.

This does not establish the opposite verdict. Sec. 13 shows the same quantity moving over 0.74 to 1.30 on the bounds when the grouping routine is changed, and the finite-shot study was run under one routine only. The conclusion available here is the weaker one: the bound-level inversion is not robust, so it cannot be used to claim that M2 beats M3 on that row.

11.3 The two BAI methods inflate differently

The inflation is not a common factor. M2 inflates by 2.2 to 3.2 on every row. M3 inflates by 2.3 to 3.0 on all four H₄ rows, where it tracks M2 closely and the M2/M3 ratio is preserved to within about 10%, but by only 1.40 to 1.66 on LiH and both H₂O rows, where the M2/M3 ratio roughly doubles. The separation is far outside the quoted standard errors.

The natural explanation is overshoot. A real run does not know the deficits Δ_i in advance, so it discovers them by shrinking a common radius and overshooting each elimination point. M3’s breadth term is a single coarse sweep of the whole pool, a fixed cost that no schedule can overshoot, whereas every one of M2’s per-generator costs is discovered by overshooting. On that reading the more breadth-dominated the system, the less of M3’s cost is exposed to the schedule.

The data support the effect but not that explanation of it. If breadth-dominance were the mechanism, the two H_4 CISD rows should behave like LiH and H_2O : Table 5 puts their M2 breadth shares at 87.1% and 57.1%, higher than H_2O stretch’s 80.1% in one case. They do not. Both show M3 inflating as much as M2, exactly like the depth-dominated H_4 HF rows. What the seven rows actually separate on is the system, not the breadth share: the three large-pool cases behave one way and all four H_4 cases the other. Pool size, context count and breadth share are confounded across four systems, and the overshoot mechanism remains a hypothesis that a designed experiment — the same system at several pool sizes, or the same pool at several schedules — has not yet been run to test.

What survives without an explanation is the observation itself: on the three largest cases here, universal contexts help substantially more under finite shots than the planning bounds suggest.

11.4 Reconciling the two directions

Sec. 9.3 argues that M3’s advantage should *decay* as the system grows; this section finds it *growing*. They are not in contradiction, because they are statements about different quantities, and it is worth being explicit about which.

The fragmentation argument is about the *planning bound*, and within it about the depth term only: as the parent support grows, each individual gradient is split over more parent contexts, so resolving the final pair through shared contexts costs more relative to resolving it through their own. That mechanism is real and is measured in Table 6. The finite-shot result is about *realised* cost under a schedule, which the bound does not model at all. The two therefore do not compete, and no mechanism is offered here for the second: the previous subsection records that the obvious one fails.

Two cautions. The realised M2/M3 ratios that drive the second trend are largest on H_2O (2.24 and 1.43), and conclusion 3 withdraws every M2-versus-M3 verdict on H_2O because the bound-level ratio moves across unity with the grouping routine; the finite-shot study was run under one routine, so those two entries inherit the same caveat and should not be used on their own. And “fading” in the conclusions refers only to the bound-level attribution of Sec. 10, where the sharing-only factor falls from 3.4 on H_4 to 1.1 on H_2O . Neither trend is established as a function of size: both rest on four systems in which size, geometry and state vary together.

11.5 The confidence convention is far stricter than δ , by an amount the data can only bound

No trial in the main finite-shot study of Table 10 selected the wrong generator, and sweeping the family-wise failure probability over nearly its whole range barely changes that. Errors do appear elsewhere — at $\delta = 0.90$ and 0.99 below, at small z_δ in the lower block of Table 12, and under correlated noise in App. C — so the statement is about this sweep, not about every run in the report.

How much stricter cannot be measured here. With 300 trials and zero errors a Wilson 95% interval puts the rate anywhere between 0 and 1.3%. The 1.2×10^{-5} quoted below is the analytic prediction of Eq. (25), not a measurement; resolving a rate that small needs of order 10^6 trials. What is established is that the realised rate is at most 1.3% where 0.05 was asked for, about four times smaller, and that it barely moves as δ is swept — enough for the practical conclusion that δ is not the knob, not enough to say by how much the convention over-delivers.

The reason is structural. Whether a run halts on one surviving arm or on the radius, the leader can only be separated from the runner-up once the common radius has come down to about $\text{gap}/2$, at which point each estimate has standard error $\text{gap}/(2z_\delta)$. The leader’s margin is one gap, so in units of the standard deviation of the difference it is $z_\delta\sqrt{2}$, and

$$P(\text{wrong selection}) \approx 2\bar{\Phi}(z_\delta\sqrt{2}), \quad (25)$$

where $\bar{\Phi}$ is the upper-tail standard normal function. The single-competitor probability is $\bar{\Phi}(z_\delta\sqrt{2})$; the factor of two is a conservative allowance, not a derivation, covering the comparison being between absolute values and more than one arm sitting near the top. Dropping it halves every predicted rate and changes no conclusion. The Bonferroni term enters only through z_δ , and the stopping radius supplies most of the protection.

Eq. (25) predicts 0.049, 0.011, 0.003 and 1.2×10^{-5} at $z_\delta = 1.39, 1.80, 2.10$ and 3.10 . It accounts only for the final comparison and ignores generators wrongly eliminated earlier in the run, so it underestimates the error at small z_δ : it predicts 5% at $z_\delta = 1.39$ where the simulation gives ten to twelve. The cost is also not monotone. Below $z_\delta \approx 1.8$ it rises again, because bad eliminations prolong the run. There is an optimum near $z_\delta \approx 1.8$ to 2.1 , worth about 1.7 in realised cost at an error rate whose 95% interval reaches 3.6% at worst — a real saving, but not

Table 12: Confidence sweeps on H₄ side 1.0 HF. Upper block: the stated δ against the realised selection-error rate, 300 trials per entry. Lower block: the confidence factor set directly, 500 trials per entry. Brackets are Wilson 95% intervals. At these trial counts none of the upper rows distinguishes a rate of 10^{-5} from one of 10^{-2} .

δ	z_δ	M2 error [95% CI]	M3 error [95% CI]	M2 shots	M3 shots
0.05	3.10	0.0% [0, 1.3]	0.0% [0, 1.3]	1,891,179	1,324,843
0.40	2.42	0.0% [0, 1.3]	0.0% [0, 1.3]	1,232,695	849,844
0.90	2.11	0.7% [0.2, 2.4]	0.3% [0.1, 1.9]	1,128,410	691,975
0.99	2.07	0.0% [0, 1.3]	0.3% [0.1, 1.9]	969,099	741,804
—	1.39	12.2% [9.6, 15.4]	10.6% [8.2, 13.6]	1,586,040	989,854
—	1.80	1.8% [1.0, 3.4]	2.0% [1.1, 3.6]	1,113,341	819,684
—	2.10	0.4% [0.1, 1.5]	0.0% [0, 0.8]	1,211,688	685,567
—	3.10	0.0% [0, 0.8]	0.0% [0, 0.8]	2,004,737	1,335,779

the factor of five the bound suggested. Both conventions are selectable in the implementation; the tables in this report retain the Bonferroni one. The sweep was run on one case, H₄ side 1.0 with the Hartree–Fock reference, and the optimum has not been checked on any other, so $z_\delta \approx 1.8$ to 2.1 should be treated as a single-system observation rather than a recommended default.

11.6 Good-enough selection

A run does not have to identify the maximiser; it is often enough to return a generator that is not much worse. Fix a tolerance $\tau > 0$ — written τ to keep it distinct from the deficits Δ_i — and stop at the first round t at which every surviving candidate is certified within τ of the best,

$$\max_{j \in \mathcal{A}_t} (|\hat{g}_j| - r_j) \geq \max_{i \in \mathcal{A}_t} (|\hat{g}_i| + r_i) - \tau, \quad (26)$$

returning $\arg \max_{i \in \mathcal{A}_t} |\hat{g}_i|$. The run is counted a success if the returned generator’s true absolute gradient is within τ of the true maximum.

On H₄ side 1.0 HF at $\tau = \text{gap}/4 = 5.38 \times 10^{-3}$, over the same 200 trials at seed 0 as Table 10, M2 falls from $1,889,539 \pm 64,528$ to $1,398,814 \pm 32,966$ shots and M3 from $1,320,830 \pm 57,944$ to $904,382 \pm 20,201$, savings of 26% and 31%. In all 200 trials the generator returned was still the exact maximiser. Note that $\tau = \text{gap}/4$ uses the exact gap, so this inherits the oracle-knowledge caveat of Sec. 9; a deployment would set τ from a chemical accuracy target instead. Measured on one case only.

12 Parameter sensitivity

12.1 What removing the two free parameters was worth

The previous revision evaluated M1 at a target radius of $\text{gap}/4$ with no task-level justification, and evaluated the BAI methods on a geometric schedule with shrink factor $\gamma = 0.5$. Both have been replaced. The M1 target is now $\text{gap}/2$, the loosest common radius at which a non-adaptive estimate separates the winner from the field, so that all three methods answer the same question; this divides the M1 column by four. The BAI methods are evaluated in the schedule-free limit; a coarse geometric schedule overshoots each elimination point and overpays.

The schedule change is not cosmetic. Under $\gamma = 0.5$, M3 cost more than M1 on three H₄ rows. In the exact limit it does not, on any row (Sec. 9), so that ordering is a property of the limit rather than of the method; under a coarse schedule, and under finite shots (Sec. 11.1), it can fail.

Table 13: Cost of the coarse $\gamma = 0.5$ schedule at the planning level, with no noise: the factor by which each method overpays against its exact limit, and what that does to their ratio. The schedule is expensive but close to neutral between the methods.

Case	M2 γ/exact	M3 γ/exact	M2/M3 exact	M2/M3 at $\gamma=0.5$	change
LiH, $R=3.0$, HF	3.80	3.53	1.79	1.92	+7%
H ₄ , side 1.0, HF	1.75	1.64	1.51	1.61	+7%
H ₄ , side 1.0, CISD	3.48	3.07	1.92	2.19	+14%
H ₄ , side 2.0, HF	2.75	2.85	1.92	1.85	−4%
H ₄ , side 2.0, CISD	2.28	1.91	2.60	3.11	+20%
H ₂ O eq., HF	3.61	3.44	1.13	1.19	+5%
H ₂ O str., HF	3.67	3.51	0.86	0.90	+5%

Refining the schedule converges onto the exact limit as it must: on H₄ side 1.0 HF, M2 falls $1,494,274 \rightarrow 1,267,755 \rightarrow 1,080,113 \rightarrow 885,115$ and M3 $928,366 \rightarrow 872,427 \rightarrow 712,219 \rightarrow 584,142$ at $\gamma = 0.5, 0.7, 0.85, 0.97$, against exact limits of 854,629 and 566,951. `verify_part1.py` asserts this convergence.

For M2 the limit is the closed form

$$M_{\text{M2}} = \sum_i \left(\frac{z_\delta \sum_\alpha \sigma_{i\alpha}}{\Delta_i/2} \right)^2, \quad (27)$$

with Δ_i replaced by the gap for the leader. For M3 the trajectory has one breakpoint per distinct Δ_i and the allocation of Eq. (21) is evaluated at each.

12.2 The confidence level does not change the comparisons

Every planning bound scales with z_δ^2 , so every ratio in Table 3 is invariant under the family-wise failure probability. Verified for $\delta \in \{0.2, 0.05, 0.01, 0.001\}$: the H_4 side 1.0, HF ratios stay at 2.63 and 1.51 to better than one part in 10^3 , while the absolute costs move by about a factor of two.

This invariance is a property of the bounds and does not extend to realised cost. Once a schedule and finite shots are introduced, z_δ changes where each elimination fires and therefore how much each method overshoots, and the two methods are not affected equally: Table 12 gives realised M2/M3 ratios of 1.60, 1.36, 1.77 and 1.50 at $z_\delta = 1.39, 1.80, 2.10$ and 3.10 on one H_4 row — a spread of 1.3, with no monotone pattern. What δ does to the realised error rate is treated in Sec. 11.5.

13 Routine independence

Every number above comes from one first-fit grouping with one insertion order. First-fit is deterministic given an order, but the order is a free convention, so the conclusions have to be tested against it. The insertion order is now a parameter: three deterministic conventions — by decreasing weight (**weight**, the default used everywhere else), by increasing weight (**reverse-weight**), and lexicographic — together with a **random** family under seeds 0, 1, 2, 3. The H_4 sweep uses all seven; the LiH sweep uses the three deterministic orderings and **random** seed 0; the H_2O sweep uses the four orderings with **random** seed 0 and varies the parent grouping only, leaving M2’s and the baseline’s individual groupings at the default. Every combination is recorded per case in `*grouping_routines.csv`.

Table 14: Spread of the method ratios across groupings of the same problem. H_4 uses seven groupings and LiH four, each recomputing the baseline and all three methods. The H_2O row varies the parent grouping only, over four orderings.

Case	Contexts	M1/M2	M2/M3	M1/M3	Baseline/M3
H_4 , side 1.0, HF	49–61	1.91–2.63	1.51–1.97	3.35–4.55	8.47–12.27
LiH, $R=3.0$, HF	980–1,127	2.58–3.12	1.79–2.11	5.22–5.58	10.68–12.68
H_2O str., $R=1.75$, HF	2,356–2,870	—	0.74–1.30	174–262	—

On H_4 and LiH the ratios move by at most a factor of 1.45 and the ordering of the methods is identical under every grouping tested. Those two systems are routine-independent for the quantities in the table.

H_2O is not. The M3 cost varies by a factor of 1.77 across parent groupings, from 1.34×10^7 to 2.37×10^7 , and the M2/M3 verdict changes sign: M3 is cheaper under two of the four orderings and more expensive under the other two. **The M2 versus M3 comparison on H_2O is therefore not resolved.** It lies inside the spread induced by an arbitrary convention, and no verdict should be claimed in either direction — neither the planning-bound result that M2 wins on the stretch HF row, nor the finite-shot result of Sec. 11 that M3 does.

This applies to all four H_2O rows, not only the swept one. The three unswept rows give M2/M3 = 1.13, 1.03 and 1.01, every one of which lies inside the measured 0.74 to 1.30 spread. They are reported in Table 3 because they are what the fixed convention gives, but none of them is far enough from unity to survive the convention being changed, and the sweep that would settle them has not been run. The M1 versus M3 comparison on the same rows is never in doubt, at 174 to 262 on the swept row and 7.8 to 192 across the four.

14 Within-context correlation

Assumption (iii) of Sec. 7 is removed here. The covariance of two generators’ fragments inside one context follows from variances alone,

$$\text{Cov}(F_{i\alpha}, F_{j\alpha}) = \frac{1}{2} [\text{Var}(F_{i\alpha} + F_{j\alpha}) - \text{Var}(F_{i\alpha}) - \text{Var}(F_{j\alpha})], \quad (28)$$

and sampling from those matrices makes the simulation exact for the shared methods as well. The matrices are formed per context, symmetrised, and factorised through an eigendecomposition with negative eigenvalues clipped to zero, which keeps the draw well defined when a context carries almost no variance.

The sampler has been verified. On all three cases below, every context covariance is positive semidefinite to rounding (most negative eigenvalue -5×10^{-16}), every diagonal entry equals the corresponding $\sigma_{i\alpha}^2$ exactly, and the empirical marginal variance of each generator over 2×10^4 draws lies between 0.976 and 1.023 of its analytic value. The anomaly reported below is therefore not a marginal-sampling defect.

14.1 The covariance-aware elimination rule, and a sign defect in it

Where the *marginal* rule of Eq. (9) compares $|\hat{g}_i|$ and $|\hat{g}_j|$ at their individual radii, the *pairwise* rule compares each pair at the radius of the combination the comparison actually depends on. Because the comparison is between

absolute values, that combination is $\widehat{g}_i - s_{ij}\widehat{g}_j$ with $s_{ij} = \text{sign}(\widehat{g}_i\widehat{g}_j)$, so i is eliminated when some active j satisfies

$$|\widehat{g}_j| - |\widehat{g}_i| > z_\delta \sqrt{V_{ij}}, \quad V_{ij} = \sum_{\alpha} \frac{1}{n_{\alpha}} \text{Var}(F_{i\alpha} - s_{ij}F_{j\alpha}), \quad (29)$$

with the same z_δ of Eq. (20). When the two fragments are correlated in the direction that helps, V_{ij} is below $\text{Var}(\widehat{g}_i) + \text{Var}(\widehat{g}_j)$ and the rule fires earlier.

The sign factor matters and was missing from the previous revision, which used $\text{Var}(\widehat{g}_i - \widehat{g}_j)$ unconditionally. That is not only the wrong combination for an opposite-sign pair; it is not even well defined, because the sign of a gradient depends on the arbitrary phase of the Hartree–Fock orbital it excites from. Only $s_{ij} \text{Cov}(\widehat{g}_i, \widehat{g}_j)$ is phase-invariant, and Eq. (29) uses exactly that. The defect is corrected in the numbers below.

Table 15: M3 under correct joint noise, 250 trials per row at seed 0, with the corrected pairwise rule. Uncertainties are one standard error; brackets are Wilson 95% intervals on the correct-selection rate.

Case	Noise model, rule	Shots	Correct [95% CI]	vs indep.
H ₄ , side 1.0, HF	independent, marginal	1,334,743 ± 51,258	100.0% [98.5, 100]	—
	correlated, marginal	1,348,697 ± 49,914	100.0% [98.5, 100]	1.01
	correlated, pairwise	757,292 ± 65,390	98.8% [96.5, 99.6]	0.57
H ₄ , side 1.0, CISD	independent, marginal	148,219 ± 1,989	100.0% [98.5, 100]	—
	correlated, marginal	146,140 ± 1,808	100.0% [98.5, 100]	0.99
	correlated, pairwise	88,995 ± 2,630	100.0% [98.5, 100]	0.60
H ₄ , side 2.0, HF	independent, marginal	544,228 ± 10,397	100.0% [98.5, 100]	—
	correlated, marginal	6,475,180 ± 793,419	75.6% [69.9, 80.5]	11.90
	correlated, pairwise	9,070,121 ± 925,919	51.2% [45.0, 57.3]	16.67

On both H₄ side 1.0 rows, modelling the correlation changes the cost by 1 to 2%, and the covariance-aware rule then cuts M3’s realised cost by 40 to 43%, at 100.0% [98.5, 100] and 98.8% [96.5, 99.6] correct selection — the second interval excludes 100%, so this is a small measured loss, not none. This is the mechanism Part II [6] proposes for the decisive stage, appearing as a direct reduction in shots rather than as the outcome of an optimisation over coefficient splitting.

H₄ side 2.0 HF behaves differently and is not explained. App. C records what is and is not known about it. Until it is settled the independent noise model of Sec. 11 remains the one used for every reported number, and the correlated results above should be treated as *unverified* on that case.

A diagnostic, defined once. For an active pair write

$$R_{ij} = \frac{\text{Var}(\widehat{g}_i - s_{ij}\widehat{g}_j)}{\text{Var}(\widehat{g}_i) + \text{Var}(\widehat{g}_j)} = 1 - \frac{2 s_{ij} \text{Cov}(\widehat{g}_i, \widehat{g}_j)}{\text{Var}(\widehat{g}_i) + \text{Var}(\widehat{g}_j)}, \quad (30)$$

the factor by which the decisive comparison is harder ($R_{ij} > 1$) or easier ($R_{ij} < 1$) than the marginal radii assume. Because s_{ij} and Cov change sign together under an orbital phase flip, R_{ij} is phase-invariant. Over the three cases it is 0.929, 1.084 and 1.168.

15 Conclusions from the current data

Table 16 gives the evidence level; items resting on one system or on bounds alone should not be generalised without the work in Sec. 17.

1. The four constructions are implemented as separate modules saving their FC groups, fragment variances and elimination histories, evaluated at one criterion with no free schedule or precision parameter. The validation gate passes on all nine cases to 4×10^{-15} Ha.
2. **On the planning bounds M3 is cheaper than M1 in every case**, by 1.1 to 192. *Observed, not proved* — the sum-of-maxima accounting of Eq. (14) makes the obvious argument incomplete — and bound-level only: on realised cost M3 loses to M1 on both H₄ CISD rows (Sec. 11.1). Every M1 column assumes M1 is handed the exact gap.
3. **M3 is cheaper than M2 on H₄ and LiH; on H₂O it is unresolved.** M3 wins eight bound-level rows of nine, but all four H₂O rows lie between 0.86 and 1.13, inside the 0.74–1.30 spread that changing the grouping routine produces. No verdict on any H₂O row, in either direction, including the finite-shot one. The sweep itself covers H₄ side 1.0 HF and LiH only.
4. **M2 is not always cheaper than M1:** seven bound-level rows of nine, three of seven on realised cost. The counterexamples are wide-gap rows where static estimation is already cheap, but LiH has a comparably wide gap and does not invert, so the gap alone is not the explanation.

5. **The two mechanisms are substitutes.** Their interaction against the no-sharing baseline is between 0.57 and 0.96 in all nine cases: elimination removes the situation sharing exploits.
6. **Elimination dominates in seven rows of nine.** Elimination alone is worth 1.27 to 252.97, sharing alone 1.11 to 3.35; sharing wins on both H_4 CISD rows. The sharing factor is smallest on all four H_2O rows, which with four systems is a correlation with system, not a trend with size, and refers to bounds only.
7. **Under finite shots M3 inflates less than M2 on the three largest cases:** M2 by 2.2 to 3.2 everywhere, M3 by 2.3 to 3.0 on H_4 but 1.40 to 1.66 on LiH and H_2O , where the M2/M3 ratio roughly doubles. Most of each inflation figure is $\gamma = 0.5$ discretisation, but the schedule moves the *ratio* by only -4 to $+20\%$ against an observed $+125\%$, so the differential is real. No mechanism is established, breadth-dominance having failed on the H_4 CISD rows, and two of the three supporting rows are H_2O , where conclusion 3 withdraws every verdict.
8. **The confidence convention does not control the selection-error rate.** Sweeping δ from 0.05 to 0.99 moves the realised rate only within 0–0.7%; 300 trials bound it below 1.3%. Inverting Eq. (25) for 5% gives 10 to 12% in simulation. An optimum sits near $z_\delta \approx 1.8$ –2.1 on the one case tested. Invariance of the ratios to z_δ holds for planning bounds only; the realised M2/M3 ratio moves over 1.36 to 1.77.
9. **Conclusions 2 to 6 are routine-independent at bound level on H_4 side 1.0 HF and LiH,** the two rows swept: ratios move by at most 1.45, no ordering changes. No other row, and neither the finite-shot nor the correlated study, is covered.
10. **Within-context correlation is a saving on two cases and unexplained on a third.** On both H_4 side 1.0 rows it changes cost by 1 to 2% and the covariance-aware rule cuts M3’s realised cost by 40 to 43% at 98.8–100% correct. On H_4 side 2.0 HF accuracy falls to 75.6% and 51.2%; sampler, sign convention, seed noise and orbital phases are all ruled out (App. C). That case is unverified and the independent model remains the reported one.
11. The physics of the previous revision is confirmed; its cost model is not reproducible and its M1/M2 ratios, 0.28 to 627, should not be used (App. A).
12. All numbers are single-step, single-state diagnostics counting shots only — not a full ADAPT trajectory, and not circuit-level cost, where M2 and the baseline need 3.5 to 6.1 times as many distinct settings (Table 1).

Table 16: Evidence behind each conclusion. “Bound” is the oracle planning model, “finite-shot” the surrogate of Sec. 11, “swept” the grouping-routine sweep of Sec. 13.

#	Claim	Bound	Finite-shot	Swept
2	$M3 < M1$	9 of 9	5 of 7	H_4 1.0 HF, LiH
3	$M3 < M2$ (resolved on H_4 , LiH)	8 of 9	7 of 7, 3 unresolved	H_4 1.0 HF, LiH
4	$M2 < M1$	7 of 9	3 of 7	H_4 1.0 HF, LiH
5	Substitutes, interaction < 1	9 of 9	—	H_4 1.0 HF, LiH
6	Elimination dominates	7 of 9	—	H_4 1.0 HF, LiH
7	M3 inflates less on large cases	—	7 cases, 200 trials	no
8	δ does not control error	—	1 case, 300–500 trials	no
10	Covariance-aware rule	—	3 cases, 250 trials, 1 unverified	no
12	Distinct-setting counts	4 systems	—	no

16 Interface with Parts II and III

Part II improves M3 by using measured variances and covariances to optimise overlapping universal operators and coefficient splitting. Sec. 9.2 gives that work a sharper target, but Sec. 9.4 also raises a caution that should be settled before it is built.

The mechanism Part II [6] proposes for the decisive stage is a pairwise-variance identity: best-arm identification only requires the pairwise differences to be resolved, and, with the sign convention of Eq. (29),

$$\text{Var}(\hat{g}_i - s_{ij}\hat{g}_j) = \sum_{\alpha} \frac{1}{n_{\alpha}} \text{Var}(F_{i\alpha} - s_{ij}F_{j\alpha}), \quad (31)$$

so the covariance that helps enters through co-measurement inside a shared context, not through the two generators sharing Pauli products. That premise was tested on the finalist pair of each system, inside the parent contexts, by comparing the cost of resolving the difference alone against the cost of resolving both generators to the same radius.

Resolving the difference is cheaper only on H_4 , by 1.11; on LiH and both H_2O rows it is more expensive, by up to 1.7. Writing S for S^{par} : resolving two generators separately costs $S_i^2 + S_j^2$ in units of ϵ^{-2} against S_{i-j}^2 for the difference, so the difference wins only when $S_{i-j} < \sqrt{S_i^2 + S_j^2}$, which for comparable arms needs roughly 30% cancellation. The measured cancellation $1 - S_{i-j}/(S_i + S_j)$ is 33% on H_4 , 17% on LiH and 7% on H_2O stretch. It

Table 17: Separating the two finalists inside the shared parent contexts: resolving only their difference against resolving each of them, both to radius gap/2. All sums are over parent contexts, so these are the S^{par} of Sec. 9.2. The two finalist gradients have the *same* sign in all four cases ($s_{ij} = +1$), so the relevant observable is $C_i - C_j$ and no sign correction applies; the one opposite-sign pair in this study, H₄ side 2.0 HF, is not among them.

Case	S_i^{par}	S_j^{par}	S_{i-j}^{par}	Resolve both	Resolve difference
H ₄ , side 1.0, HF	1.921	1.731	2.455	554,857	499,910
LiH, $R=3.0$, HF	2.170	2.180	3.623	48,712	67,577
H ₂ O eq., $R=0.9572$, HF	7.263	12.173	17.803	1,002,360	1,581,034
H ₂ O str., $R=1.75$, HF	6.239	7.794	13.005	10,636,302	18,049,318

does not track the finalist overlaps of Table 7 — H₄ has zero overlap and the largest cancellation, LiH 35% overlap and half the cancellation — consistent with the premise that the useful covariance comes from co-measurement rather than shared support, but leaving the size of the cancellation unexplained.

This does not refute Part II, whose other mechanisms — coefficient splitting over an overlapping cover, empirically estimated covariances, adaptive choice of the next context — are untouched. It does mean the pairwise-difference mechanism should not be assumed to pay, and that cancellation must be measured on the intended benchmark first. A second caution: that mechanism acts on the depth term, only 12% of M3’s cost on LiH and 39% on H₂O stretch, so its effect is bounded even where it pays.

Part III [7] changes the ADAPT trajectory so that the commutators first become symmetry-filtered and tapered. When Part III is active, the Part I object C_i is replaced by

$$C_{i,\text{tap}}^{\parallel,k} = \text{Taper}_{\mathcal{R}_k}(\Pi_{\mathcal{R}_k}([H, G_i])), \quad (32)$$

and then M1, M2, or M3 is applied to these reduced operators. Part I tables do not include this reduction.

17 Remaining work in Part I

1. **Diagnose the H₄ side 2.0 correlated-noise failure** (App. C). Sampler, sign convention, seed noise and orbital phases are ruled out; the degenerate runner-up and the anticorrelated leading pair are the live candidates. Until it is settled the correlated model cannot be used for reported numbers.
2. **Extend the routine sweep.** Three H₂O rows, three H₄ rows, the individual groupings M2 and the baseline depend on, and the finite-shot and correlated studies are all unswept. The single largest gap in the evidence base.
3. **Separate schedule from noise** by running the surrogate across several γ , which would also show how much a tuned schedule recovers.
4. **Test the confidence convention and good-enough stopping beyond H₄ side 1.0 HF** before adopting either; the good-enough tolerance also needs defining from a chemical accuracy target rather than the exact gap.
5. **Simulate the two H₂O CISD rows**, the only ones missing from Table 10 and the ones with the largest bounds.
6. **Price the excluded cost components.** Distinct settings differ by 3.5 to 6.1 times between the method families (Table 1), so the M2-versus-M3 comparison is incomplete even where the shot counts are resolved.
7. **Complete the test suite** (Sec. 18); the highest-value gaps are brute-force commutator and variance checks, an adversarial search against $M_{\text{M3}} \leq M_{\text{M1}}$, and shot-level sampling of the surrogate.
8. **Establish whether the CISD state is a fair proxy** for a mid-trajectory ADAPT state, which needs trajectory data.

18 What is tested, and what is not

Covered. 26 unit tests pin the conventions that were previously wrong and the pieces with closed forms: basis ordering and the big-endian convention; the qubit Hamiltonian against the PySCF RHF and FCI energies on every case, as a hard gate; the top gradient against a finite-difference energy derivative; the allocation against Eq. (18) for a single gradient; and the fragment decomposition summing back to the gradient operator. `verify_part1.py` adds 68 checks on the published results: each reported total is the sum of the file that explains it, the gate passed, the allocation meets its confidence target, $M_{\text{M3}} \leq M_{\text{M1}}$, all methods agree on the selected generator, repeated runs agree exactly, the ratios are invariant to δ , and a geometric schedule converges onto the exact limit. All pass. The correlated sampler is checked separately for positive semidefiniteness, exact diagonals and empirical marginals (App. C).

Not covered, in rough order of value: term-by-term brute-force checks of the commutators, of *all* gradients against finite differences and of the fragment variances against dense $\langle F^2 \rangle - \langle F \rangle^2$, the pipeline being validated only at its ends; grouping validity (pairwise commutation inside a context, exact partition of $\mathcal{B}_0$, validity of the restricted contexts $\mathcal{M}_\alpha \cap \mathcal{B}_t$), all asserted by construction; shot-level sampling of measurement outcomes, against which neither the Gaussian marginals (assumption (i) of Sec. 7) nor the joint draw has been compared — this is what would settle App. C; an adversarial search for a counterexample to $M_{M3} \leq M_{M1}$ under the sum-of-maxima aggregation, which would turn conclusion 2 into a bounded claim; KKT residuals for the allocation on systematic multi-constraint instances; analytic cases (minimal-basis H_2 by hand, a two-arm synthetic problem pinning the elimination order and Eq. (27)); limiting and symmetry cases (a one-generator pool giving $M_1 = M_2 = M_3$, divergence as $1/\text{gap}^2$, symmetry-zero gradients below threshold, invariance under relabelling qubits or permuting the pool, pool sizes against the combinatorial formula, anti-Hermiticity); a phase-invariance test (flip one orbital’s sign and confirm every reported number is unchanged); and regression pinning of every table entry against a commit hash.

One reproducibility limit, and what it does not explain. The Hartree–Fock orbitals of square H_4 are fixed only up to the sign of each molecular orbital, and different SCF starting guesses return different signs; the orbital energies are non-degenerate at both side lengths, so the solution is unique up to those phases. Gradients and covariances always come from the same SCF run, and a phase flip changes g_i and the matching row and column of the covariance together, leaving $|g_i|$, every fragment variance and $s_{ij} \text{Cov}(\hat{g}_i, \hat{g}_j)$ — hence R_{ij} of Eq. (30) — invariant. The reported statistics are therefore phase-invariant in principle, and phases explain no difference between runs. What they prevent is comparing *signed* gradients or raw covariances across runs; a convention such as making each orbital’s largest-magnitude coefficient positive would remove that, and the invariance itself is the untested item listed above.

The implementation should be regarded as validated at the level these tests reach and no further. Two defects of the previous revision were silent and were caught only by end-to-end energy comparison (App. B).

19 Reproduction

```
python -m pip install -r requirements.txt
python scripts/run_part1_methods.py --all --out runs
python scripts/run_finite_shot.py --case H4_square_eq_side1p0_HF \
    --trials 200 --out runs
python -m pytest tests -q
python scripts/verify_part1.py --runs runs
```

Each case writes a per-method summary together with the FC groups, fragment standard deviations and BAI histories behind it, and records the interpreter and package versions with every run. `verify_part1.py` checks the published results rather than the implementation: for every case it confirms that the reported total is the sum of the file that claims to explain it, that the validation gate passed, that the allocation meets its confidence target, that $M_{M3} \leq M_{M1}$, and that all methods agree on the selected generator; against the live code it confirms that the fragment decomposition sums back to the gradient, that repeated runs agree exactly, that the ratios are invariant to the confidence level, and that a geometric radius schedule converges onto the exact limit used in the tables. All 68 checks pass, alongside 26 unit tests. Sec. 18 says what those tests do and do not cover.

Case identifiers accepted by `--case`:

LiH_R3p0_HF	
H4_square_eq_side1p0_HF	H4_square_eq_side1p0_CISD
H4_square_stretch_side2p0_HF	H4_square_stretch_side2p0_CISD
H20_eq_HF	H20_eq_CISD
H20_stretch_HF	H20_stretch_CISD

Determinism and seeds. The planning bounds contain no randomness and re-run exactly; `verify_part1.py` asserts this. The finite-shot tables use `numpy` default-RNG streams at seed 0 (Tables 10 and 15) and per-entry seeds recorded in the run summaries (Table 12). The routine sweep uses the `random` orderings at seeds 0 to 3.

Numerical settings. PySCF defaults are used for the SCF convergence threshold and initial guess, except that square H_4 needs the phase convention noted in Sec. 18 to make covariances reproducible. The CISD state is the lowest eigenpair of the Hamiltonian restricted to the ≤ 2 -excitation space, taken with a dense symmetric eigensolver below 2^{12} basis states and with `scipy.sparse.linalg.eigsh` at tolerance 10^{-12} above it. The convex-solver cross-check of Sec. 5 used SLSQP on randomly generated instances with up to eight arms and twenty contexts, agreeing to better than one part in 10^6 .

Environment and provenance. Code and run outputs live at https://github.com/Zephrou5747/ADAPT_Sampling, commit `fe4ea2e`, which holds every table entry together with the FC groups, fragment standard deviations and BAI histories behind it; each run also records its own versions. Environment: Python 3.10.12, PySCF 2.14.0, OpenFermion 1.8.1, NumPy 2.2.6, SciPy 1.15.3, one desktop core. `requirements.txt` pins NumPy ≥ 1.26 , pandas ≥ 2.0 and matplotlib ≥ 3.8 for cached reproduction, and PySCF ≥ 2.6 , OpenFermion ≥ 1.6 , NetworkX ≥ 3.0 and SciPy ≥ 1.11 for regeneration from geometry. Wall clock is dominated by building the gradient problem: about 16 s for H₄, 37 s for LiH, two to three minutes for H₂O; each 200-trial run then takes 0.1 to 40 s per method.

A Revision history

What changed against the previous revision. All numbers come from a single from-geometry implementation, one module per method, each saving the FC groups and fragment variances behind every entry, and every run passing the validation gate of Sec. 6. Both free parameters were removed: M1 is evaluated at the radius matching the winner-identification task, and the best-arm methods in the schedule-free exact limit. Added since: the no-sharing baseline (Sec. 10), the finite-shot surrogate (Sec. 11), the routine sweep (Sec. 13), the correlated noise model (Sec. 14) and the realised-cost comparison against M1 (Sec. 11.1).

Regenerated tables. The finite-shot study now uses 200 trials at seed 0 on every row, where H₂O previously used 40, with standard errors throughout; totals move by the few per cent of run-to-run scatter (LiH M3, 882,018 before against $893,509 \pm 19,804$ now). The correlated-noise table was regenerated after the sign correction of Eq. (29). The breadth metric of Table 5 changed definition, from the share paid by the top two arms to Eq. (23): on H₄ side 1.0 HF that moves M2 from 53.8% to 3.0%.

The physics is confirmed; the cost model is not reproducible. Universal Pauli supports, gradient values, gaps and selected generators agree with the previous revision to six or more digits. The present code was written from the geometry upward and shares no module with the old scripts, but it was written by the same project against the same conventions, so this is evidence that the physics layer is correct rather than that two independent groups agree. Three things changed the cost model.

The grouping. All rows now use one deterministic greedy routine; the previous H₄ and H₂O rows used a different partitioning giving 356 and 9,874 contexts where greedy gives 53 and 2,366. Cost grows with the number of fragments an operator is split across, so this alone is a factor of several.

The previous M1 column was not a single model. For LiH it is reproduced within 8% by a uniform per-context allocation sized to a *typical* rather than worst-case generator (1,929,181 against the reported 2,088,380), which guarantees the target precision for about half the gradient vector. For H₄ no combination of grouping, allocation and confidence convention reproduces the reported 182,282,619; uniform allocation at worst-case variance with the old 356 contexts lands near 3×10^7 . That entry also exceeds its own no-sharing baseline of 11,326,894 by a factor of 16. That is not strictly impossible — with $f_i > 1$ a shared allocation can exceed the baseline, bounded by $\max_i f_i$ times it — but the measured f_i are between 1.07 and 3.62 and the present M1 is below its baseline on all nine rows, so a factor of 16 remains unexplained.

The allocation. On LiH, solving Eq. (21) costs about 17 times fewer shots than spreading the same total uniformly over the 980 parent contexts, at equal ϵ and $z_\delta = 1$. At one standard error the present M1 is *cheaper* than the previously reported value, 1,191,917 against 2,088,380, despite covering the worst case. The larger absolute values in Table 3 come from the Bonferroni z_δ , applied identically to all four constructions and therefore invisible in the ratios.

The previous M1/M2 ratios, from 0.28 to 627, should not be used: they combine an optimistic LiH entry with an unexplained H₄ one.

B Naming, implementation status and past defects

The method names **M1: FC-UG**, **M2: BAI-FC-IG** and **M3: BAI-FC-UG** are used consistently in text, code, tables, plots and file names; output files carry the codes `M1_FCUG`, `M2_BAIFCIG`, `M3_BAIFCUG` and `NOSHARE`. Every number in every table is saved with its FC groups, fragment standard deviations and BAI history, and every run records its grouping conventions, interpreter and package versions. M3 is complete for H₂O, so the headline comparison has no missing entries.

Two defects of the previous revision are worth recording because both were silent. The two-electron integrals were inserted into the spin-orbital tensor in chemist rather than physicist ordering, with spin pattern (a, b, a, b) instead of (a, b, b, a) and no one-half prefactor; and the computational basis was indexed little-endian where the sparse-operator convention is big-endian. For H₄ the resulting Hartree–Fock energy was -5.21 Ha against an RHF value of -1.76 Ha. The validation gate of Sec. 6 exists to make a repeat impossible.

C The H_4 side 2.0 HF correlated-noise anomaly

Under correct joint noise this one case degrades sharply: correct selection falls from 100.0% to 75.6% [69.9, 80.5] under the marginal rule and 51.2% [45.0, 57.3] under the pairwise rule, with cost rising from 544,228 to 6,475,180 and 9,070,121 shots. Here is what has been ruled out and what has not.

Not the sampler: every context covariance is positive semidefinite to rounding (most negative eigenvalue -5×10^{-16}), every diagonal equals $\sigma_{i\alpha}^2$ exactly, and empirical marginal variances over 2×10^4 draws lie within 0.976–1.023 of analytic, here and on the two benign cases. *Not the sign convention*: the pairwise rule previously used $\text{Var}(\hat{g}_i - \hat{g}_j)$ regardless of sign, and Eq. (29) corrects it, which improves the benign cases slightly, does not recover this one, and does not touch the marginal rule, which degrades too. *Not seed noise*: the committed driver gives 6,475,180, 6,549,558 and 6,410,482 shots at 75.6, 80.4 and 76.4% correct over seeds 0, 1, 2; an earlier unversioned script reported 3,369,764 shots at 79.2%, agreeing on accuracy but not cost, and cannot be re-run, so the figures here supersede it. *Not orbital phases*: gradients and covariances come from one SCF run and the statistics are phase-invariant (Sec. 18).

Three things distinguish the case, and any explanation has to use them:

- **The runner-up is exactly degenerate**: $D \ 0,3 \rightarrow 4,7$ and $D \ 1,2 \rightarrow 5,6$ both at $|g| = 0.298387$, so there is no unique second arm and $|\mathcal{A}| = 3$.
- **The decisive comparison is harder than the marginal radii assume**: $R_{ij} = 1.168$ by Eq. (30), against 0.929 on H_4 side 1.0 HF. The previous revision reported 0.82 here and concluded the opposite; that number combined a covariance from one SCF run with gradient signs from another, and is withdrawn.
- **The gap is narrow relative to the field**: $\text{gap}/|g_{(1)}| = 5\%$ with $\max_i \Delta_i/\text{gap} = 18.9$, so a 17% error in the assumed variance of the decisive comparison has room to matter. H_4 side 1.0 CISD also has $R_{ij} > 1$, at 1.084, but its gap is 83% of $|g_{(1)}|$ and nothing goes wrong.

The natural reading is that an anticorrelated, near-degenerate leading pair makes the marginal radii anti-conservative, so arms are eliminated on weaker evidence than the rule believes. That fits all three observations but is not demonstrated, and it does not explain why the *pairwise* rule, which uses R_{ij} explicitly, is worse rather than better.

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